Akash Sankar Vigneshwaran

M.Sc. Physics | Astrophysics research
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Generated from the portfolio, not the original CV. Request the original by email or on the contact page.

Research interests: dark matter, cosmology, galaxy dynamics, radio astronomy instrumentation, and scientific data analysis.

EDUCATION

M.Sc. Physics - University of Bremen
Oct 2026 - Present | Bremen, Germany | In progress

B.Sc. Physics - Government Arts College, Udhagamandalam
Aug 2022 - May 2025 | Bharathiar University | CGPA 8.35 / 10

SELECTED RESEARCH AND APPLICATION DEVELOPMENT

Exoplanet Explorer
Jul - Aug 2026 | Developer & data analyst
Built NASA data ingestion, habitability scoring, and interactive views. A live dashboard with planet comparisons and CSV export.

Dark Matter & Galaxy Rotation Curves
Jun - Jul 2025 | Researcher
Reviewed rotation curves, particle candidates, and alternative gravity. A theoretical synthesis of the evidence and open questions.

Astronomy, Astrophysics & Cosmology - Literature Review
Jan - Apr 2025 | Researcher & writer
Read, compared, and synthesized primary research literature. A formal research paper grounded in a structured literature review.

RESEARCH AND INSTRUMENTATION EXPERIENCE

R&D Intern - Radio Astronomy Instrumentation - King Electricals and Engineering Works
Aug 2025 - Apr 2026
Development and testing of instrumentation for a radio telescope array, including low-noise RF enclosure design and EMI mitigation.

Training & Internship on Astrophysics - IIT Madras with Edutech Life
Feb - Mar 2025
Internship project on the history and evolution of black hole theories, with hands-on work in the computational tools used to simulate celestial phenomena.

Training & Internship on Astrophysics - IIT Bhubaneswar with Edufabrica Pvt. Ltd.
Mar - Apr 2023
Research on black holes and their role in astrophysics, reviewing the literature and reporting the theory behind it.

CONFERENCE PRESENTATION

Exoplanets and the Search for Alien Life
Oral presentation | 44th Meeting of the Astronomical Society of India | IIT Guwahati, India | May 2026

TECHNICAL SKILLS AND LANGUAGES

Tools: Python (Astropy, NumPy, Matplotlib, SciPy); SQL (Data querying and management); Advanced Excel (Data processing and visualization)
Languages: Tamil - Mother tongue; Telugu - Mother tongue; English - B2 - independent user
Memberships: ASTRAE Club, IISc Bangalore; All India Astronomy Club